

a glimpse of
typst

Plan

1. Introduction to Typst

Bibliography

Introduction

boxes

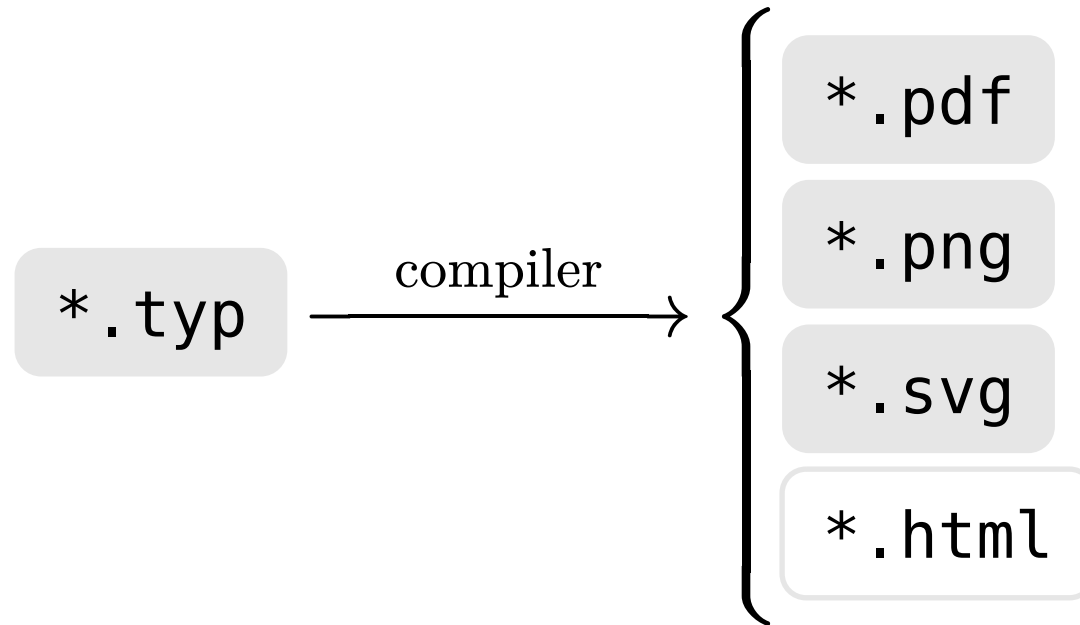
2D input in math mode

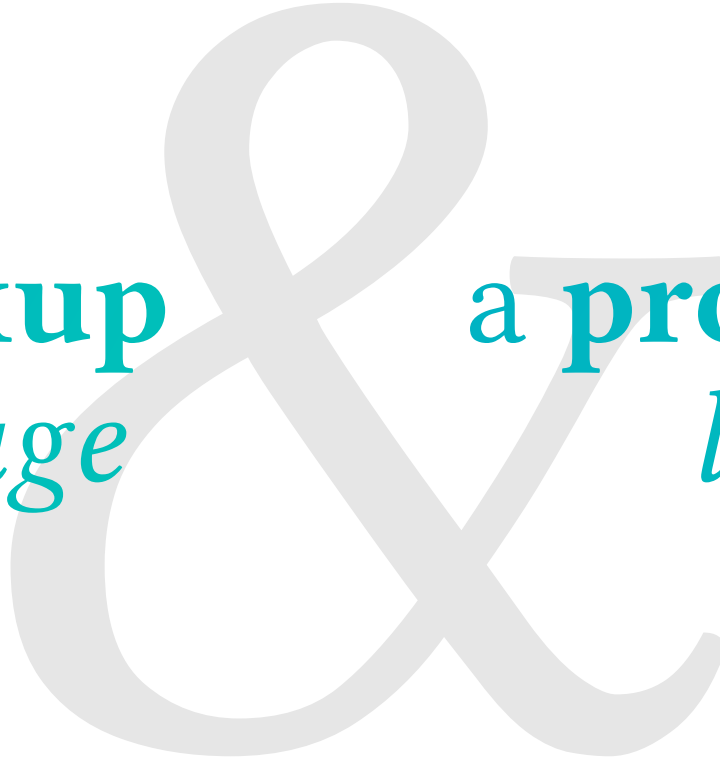
metadata

WASM plugins

2. In practice: comparison with LaTeX

Typst is a
typesetting
system





a markup
language

a programming
language

The project in numbers (as of April 2026)

Started by 2 students at TU Berlin, currently a team of 5.

Command-line compiler

- Open source release in 2023
- 414 contributors
 - 20 in last month

Community and ecosystem

- 2934 users on the forum
- 1269 packages

Webapp

- 200K users
- 600K projects

Funding

- WebApp (Pro plan and On-Premises)
- One engineer funded by Posit (Quarto)
- Initial €225k angel + EU grants

Webapp

Try it at typst.app/play/

Free plan

✓ Live preview

✓ Real-time collaboration

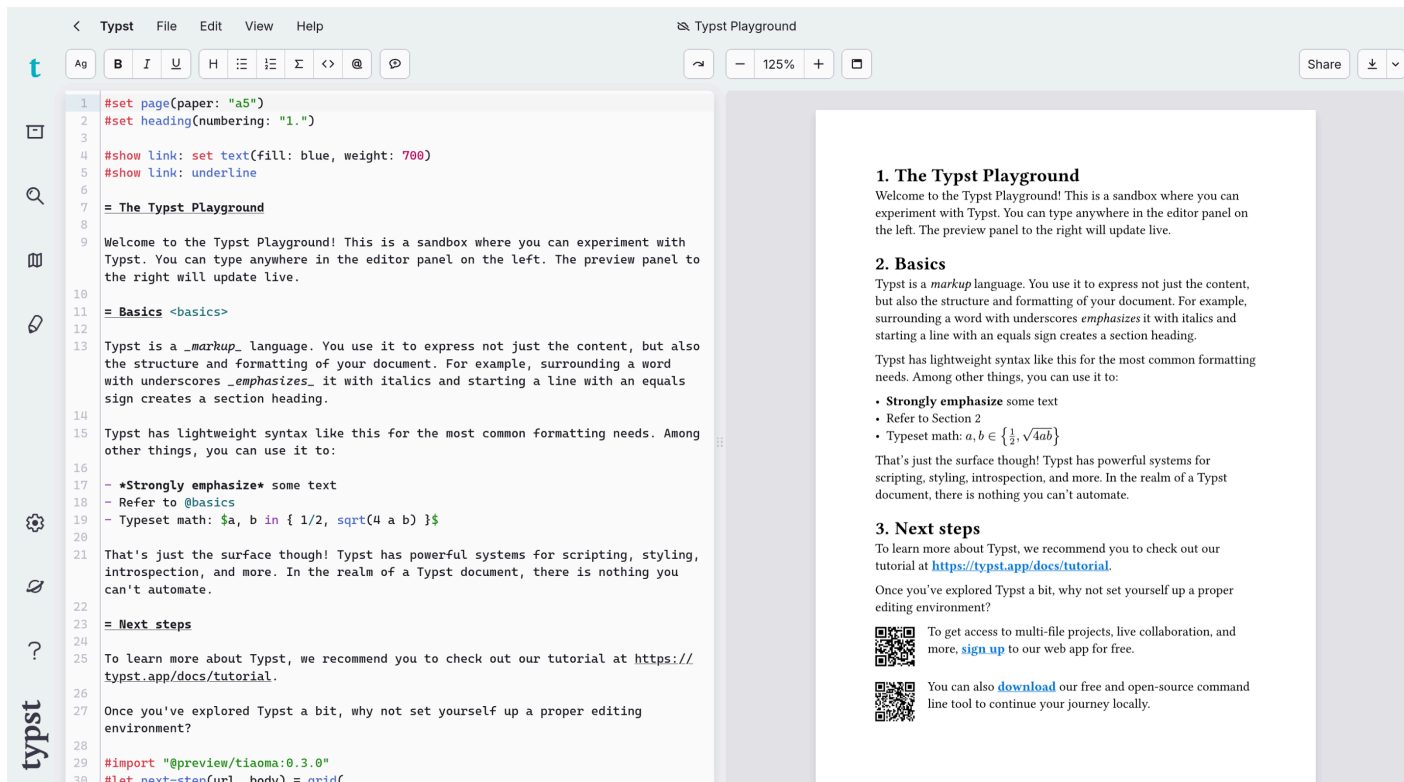
Pro plan

✓ Comments

✓ GitHub/GitLab sync

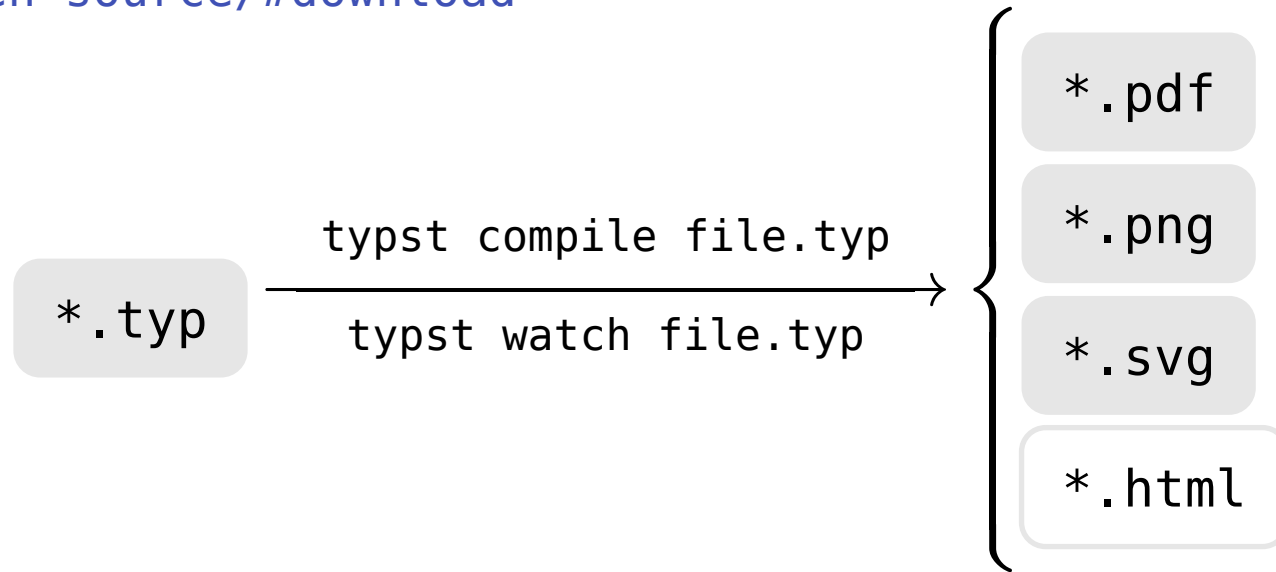
✓ Zotero/Mendeley sync

✓ Presentation mode (pointer, pen, speaker mode...)



Local installation

typst.app/open-source/#download



Typst is a
markup
language

3 modes:
markup, *math*, code

Markup mode

```
1 = Introduction
2 Some things are *important* and
3 must be _emphasized_.
4
5 == First approach <part1>
6 Some more text.
7
8 == Second approach
9 Something different from @part1.
10
```

1. Introduction

Some things are **important** and must be *emphasized*.

1.1. First approach

Some more text.

1.2. Second approach

Something different from Section 1.1.

1 | Enumerations:

2 | + Lemon

3 | + Banana

1 | + Lemon

2 |

3 | + Banana

1 | Three italian dishes are:

2 | - Pasta

3 | - Pizza

4 | - Tiramisù

Enumerations:

1. Lemon

2. Banana

1. Lemon

2. Banana

Three italian dishes are:

• Pasta

• Pizza

• Tiramisù

```
1 | We can write code inline
2 | like this `print("Hello")` or
3 | like this ```py print("Hello")```.
```

We can write code inline like this
`print("Hello")` or like this `print("Hello")`.

```
1 | We can make a code block like this:
2 | ```py
3 | print("Hello")
4 | ```
```

We can make a code block like this:
`print("Hello")`

```
1 | // We can make a comment
2 | /* In two different ways */
```

Math mode

```
1 Euler's formula:
2 $
3   e^(i theta)
4   = sum_(n=0)^oo (i theta)^n / n!
5   = cos(theta) + i sin(theta).
6 $
```

```
1 Using $theta = pi$ the formula
2 gives $e^(i pi) = -1$.
```

Euler's formula:

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} = \cos(\theta) + i \sin(\theta).$$

Using $\theta = \pi$ the formula gives $e^{i\pi} = -1$.

```

1 | $
2 |   A inter B = emptyset =>
3 |   A without B = A
4 | $

```

$$A \cap B = \emptyset \Rightarrow A \setminus B = A$$

```

1 | $
2 |   Q = 2 pi E_"start" /
3 |       E_"dissipated"
4 | $

```

$$Q = 2\pi \frac{E_{\text{start}}}{E_{\text{dissipated}}}$$

```

1 | $ W = arrow(F) dot arrow(d) $

```

$$W = \vec{F} \cdot \vec{d}$$

```
1 | $ QQ = { p/q : p in ZZ, q in NN } $
```

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\}$$

```
1 | $
2 |   vec(3,9,15) =
3 |   mat(1,2,3; 4,5,6; 7,8,9)
4 |   vec(1,1,0)
5 | $
```

$$\begin{pmatrix} 3 \\ 9 \\ 15 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

```
1 | $
2 |   f(x) &= (x+1)^2 &&= x^2 + 2x + 1 \
3 |   g(x) &= 10(x-1) &&= 10x - 10
4 | $
```

$$f(x) = (x + 1)^2 = x^2 + 2x + 1$$

$$g(x) = 10(x - 1) = 10x - 10$$

Code mode

```
1 | Example: bold text
2 |
3 | - With *markup*
4 |
5 | - With #strong[function call]
```

Here [] goes back to markup mode. Compare:

```
1 | - #strong[Some _markup_ value]
2 | - #strong("Some _string_ value")
```

Example: bold text

- With **markup**
- With **function call**

- Some *markup* value
- Some *_string_* value

```
1 #let n = 3
2
3 Powers of 2 up to $n = #n$:
4
5 #for i in range(n+1) {
6     let value = calc.pow(2, i)
7     $ 2^#i = #value $
8 }
```

Powers of 2 up to $n = 3$:

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

```
1 | #bibliography(  
2 |     "example.bib",  
3 |     full: true  
4 | )
```

Bibliography

- [1] M. A. N. & Isaac L. Chuang, *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.
- [2] P. W. Shor, "Scheme for reducing decoherence in quantum computer memory," *Physical review A*, vol. 52, no. 4, p. R2493, 1995.

```
1  #{
2    let N = 10
3    let n = 1
4    [1 ]
5    for i in range(2, N + 1) {
6        n += i
7        [\+ #i ]
8    }
9    [\= #n]
10 }
```

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

```
1 | #let nice(it) = [✨] + it + [✨]  
2 | #nice([Typst]) \ #nice[Typst]
```

✨Typst✨
✨Typst✨

```
1 | #link("example.com",[_Example_])  
2 | #link("example.com")[_Example_]
```

Example Example

```
1 | #figure(  
2 |   image("images/cow.jpg", width:  
3 |     6cm),  
4 |   caption: [Cowdynamics],  
5 | )
```

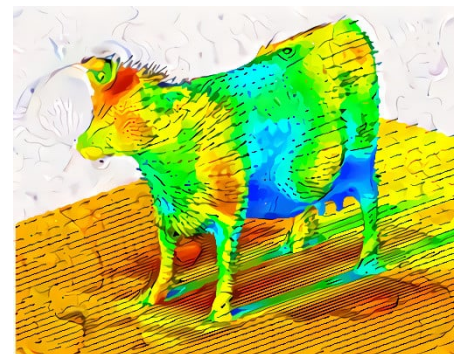


Figure 1: Cowdynamics

set & show

element
#set text(size: 30pt)
new defaults

```
1 #set text(  
2   font: "New Computer Modern",  
3   size: 24pt,  
4 )  
5  
6 This is the familiar LaTeX font.
```

This is the familiar LaTeX font.

```
1 #set enum(numbering: "a1")  
2 + Element A  
3   + Element A1  
4   + Element A2  
5 + Element B
```

- a) Element A
 - 1) Element A1
 - 2) Element A2
- b) Element B

selector
#show link: set text(blue)

#show link: underline
transformation

```
1 | #show link: set text(red)
2 | #show link: underline
3 | Go to https://www.example.com.
```

Go to <https://www.example.com>.

```
1 | // Anonymous transformation function
2 | #show link: it => [] + it
3 | Go to https://www.example.com.
```

Go to  <https://www.example.com>.

```

1 #set text(font: "Pennstander")
2
3 #show math.equation: set text(
4   font: "Pennstander Math",
5 )
6
7 The definite integral of  $f$ 
8 on  $[a, b]$  is defined as
9
10 $
11   integral_a^b = lim_(n->oo)
12     (b-a)/n sum_(i=1)^n f(xi_i)
13 $

```

The definite integral of f on $[a, b]$ is defined as

$$\int_a^b = \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{i=1}^n f(\xi_i)$$

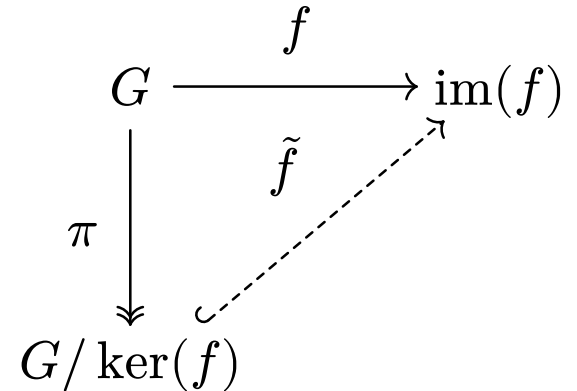


Packages

from the Typst Universe

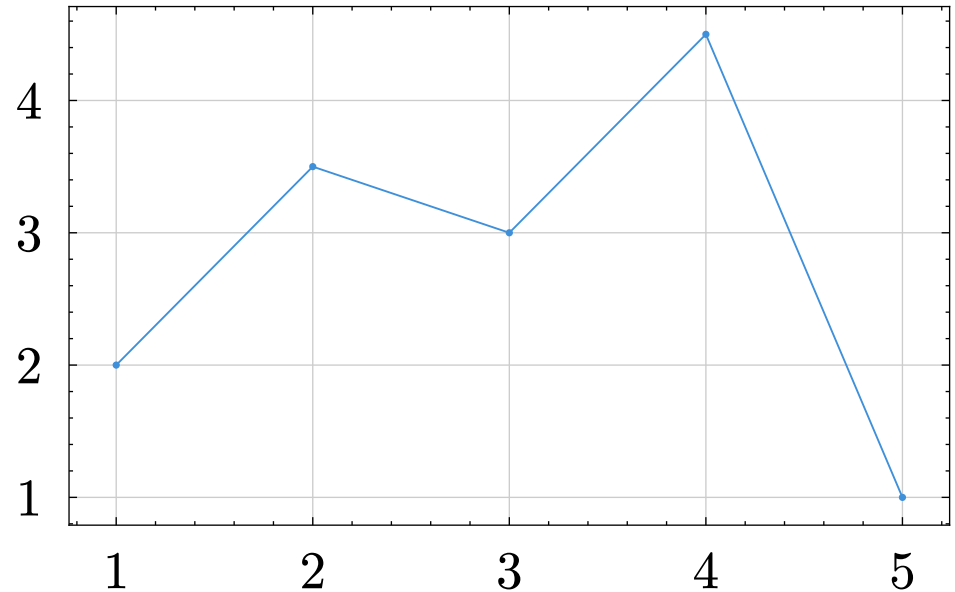
Diagrams with fletcher

```
1 #import "@preview/fletcher:0.5.8": *
2
3 #diagram(cell-size: 15mm, $
4   G edge(f, ->) edge("d", pi, ->>) & im(f) \
5   G slash ker(f) edge("ur", tilde(f), "hook-->")
6 $)
```



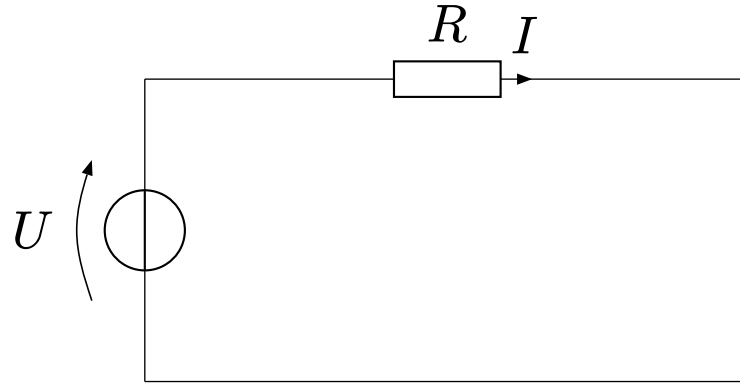
Plotting from CSV with Lilaq

```
1 #import "@preview/lilaq:0.6.0"  
2  
3 #let data = lilaq.load-txt(  
4   read("data.csv"), header: true)  
5  
6 #lilaq.diagram(  
7   lilaq.plot(data.t, data.U),  
8   width: 100% - 1cm,  
9   height: 60%,  
10 )
```



Electronic circuits with Zap

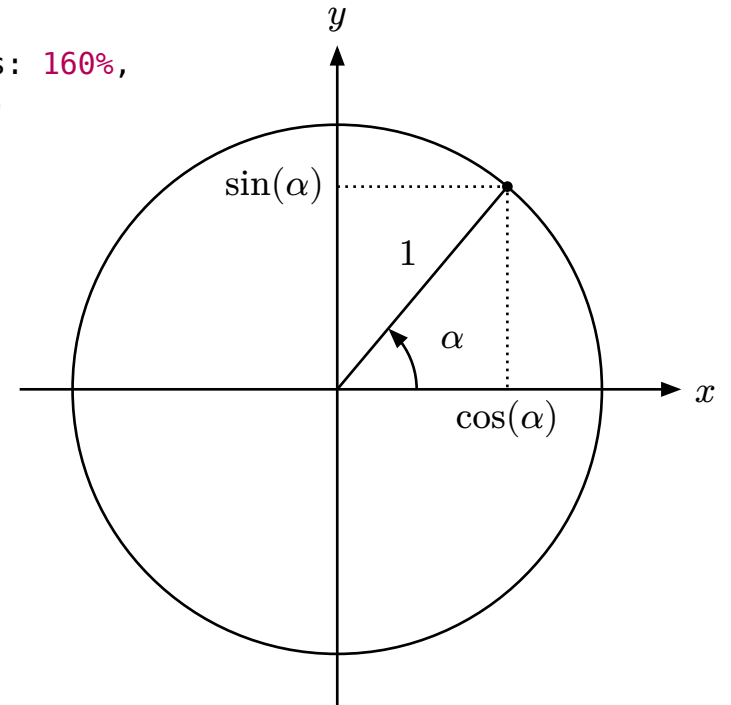
```
1  #import "@preview/zap:0.5.0"
2
3  #zap.circuit({
4      import zap: *
5
6      vsource("v1", (0, 0), (0, 4),
7          u: $U$)
8      resistor("r1", "v1.out", (8, 4),
9          label: $R$, i: $I$)
10     wire((8, 4), (8, 0))
11     wire((8, 0), (0, 0))
12 })
```



General-purpose drawing with CeTZ

```
1 #import "@preview/cetz:0.4.2": canvas, draw, angle
2 #canvas(length: 3.5cm, {
3   import draw: *
4   let my-alpha = 50deg
5   let arrow = (end: ">", fill: black)
6   let node = content.with(padding: 5pt)
7   let P = (my-alpha, 1)
8   let Px = (P, "|-", (0, 0))
9   let Py = (P, "-|", (0, 0))
10  line((-1.2, 0), (1.3, 0), mark: arrow)
11  node((), anchor: "west", $x$)
12  line((0, -1.2), (0, 1.3), mark: arrow)
13  node((), anchor: "south", $y$)
14  circle((0, 0), radius: 1)
15  circle(P, radius: 1.5pt, fill: black)
16  line((0, 0), P)
17  line(Py, P, Px, stroke: (dash: "dotted"))
18  node(Px, anchor: "north", $cos(alpha)$)
19  node(Py, anchor: "east", $sin(alpha)$)
```

```
20 node((my-alpha, 0.55), anchor: "south-east", $1$)
21 angle.angle(
22   (0, 0), (1, 0), P,
23   label: $alpha$,
24   radius: 0.3,
25   label-radius: 160%,
26   mark: arrow)
27 }
```

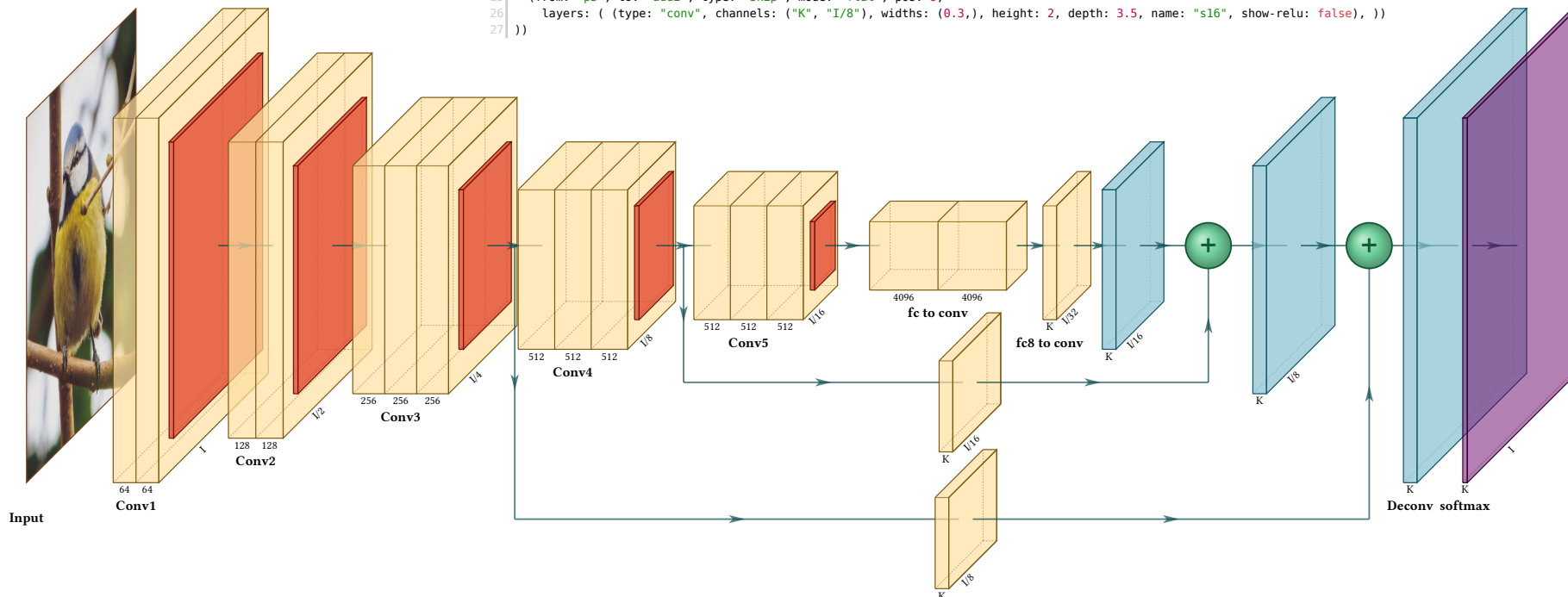


Neural network architectures with neural-netz

```

1 #import "@preview/neural-netz:0.3.0": draw-network
2 #draw-network((
3   ns: input, image: "default", height: 8, depth: 8, label: "input", name: "img"),
4   (type: "pool", height: 5, depth: 5, name: "p1"),
5   (type: "conv", channels: ("256", "256", "256", "I/4"), widths: (0.7, 0.7, 0.7), height: 5, depth: 5, label: "Conv3", name: "c3"),
6   (type: "pool", height: 3.5, depth: 3.5, name: "p3"),
7   (type: "conv", channels: ("512", "512", "512", "I/8"), widths: (0.8, 0.8, 0.8), height: 3.5, depth: 3.5, label: "Conv4", name: "c4"),
8   (type: "pool", height: 2.5, depth: 2.5, name: "p4"),
9   (type: "conv", channels: ("512", "512", "512", "I/16"), widths: (0.8, 0.8, 0.8), height: 2.5, depth: 2.5, label: "Conv5", name: "c5"),
10  (type: "pool", height: 1.5, depth: 1.5, name: "p5"),
11  (type: "conv", channels: ("4096", "4096"), widths: (1.5, 1.5), height: 1.5, depth: 1.5, label: "fc to conv", name: "fc"),
12  (type: "conv", channels: ("K", "I/32"), widths: (0.3, ), height: 2.5, depth: 2.5, label: "fc8 to conv", name: "s32", offset: 0.8, show-relu: false),
13  (type: "deconv", channels: ("K", "I/16"), height: 3.5, depth: 3.5, name: "up1", offset: 1),
14  (type: "sum", radius: 0.5, symbol: "+", name: "add1", offset: 1),
15  (type: "deconv", height: 5, depth: 5, channels: ("K", "I/8"), name: "up2", offset: 0.5),
16  (type: "sum", radius: 0.5, symbol: "+", name: "add2", offset: 1),
17  (type: "deconv", height: 8, depth: 8, channels: ("K", ), label: "Deconv", name: "up3", offset: 0.5),
18  (type: "convsoftmax", height: 8, depth: 8, channels: ("K", "I"), label: "softmax", offset: 1),
19 ), connections: (
20   (from: "p4", to: "add1", type: "skip", mode: "flat", pos: 3,
21     layers: ( (type: "conv", channels: ("K", "I/16"), widths: (0.3, ), height: 2, depth: 3.5, name: "s16", show-relu: false), ) ),
22   (from: "p3", to: "add2", type: "skip", mode: "flat", pos: 6,
23     layers: ( (type: "conv", channels: ("K", "I/8"), widths: (0.3, ), height: 2, depth: 3.5, name: "s16", show-relu: false), ) )
24 ))

```



```
#import "@preview/physica:0.9.8": *
```

```
1 $  
2 curl (grad V + pdv(vb(A), t)) =  
3 pdv(vb(B), t)  
4 $
```

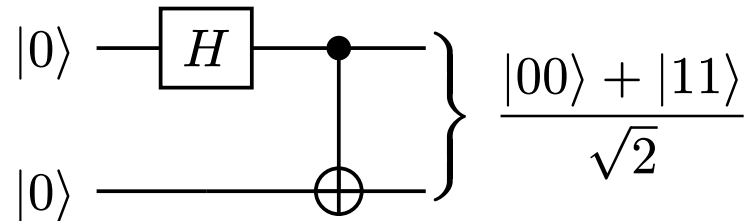
$$\nabla \times \left(\nabla V + \frac{\partial A}{\partial t} \right) = \frac{\partial B}{\partial t}$$

```
1 #let tn = tensor  
2 $  
3 tn(R, +mu, -nu) -  
4 1/2 R tn(g, +mu, -nu) =  
5 8 pi G tn(T, +mu, -nu)  
6 $
```

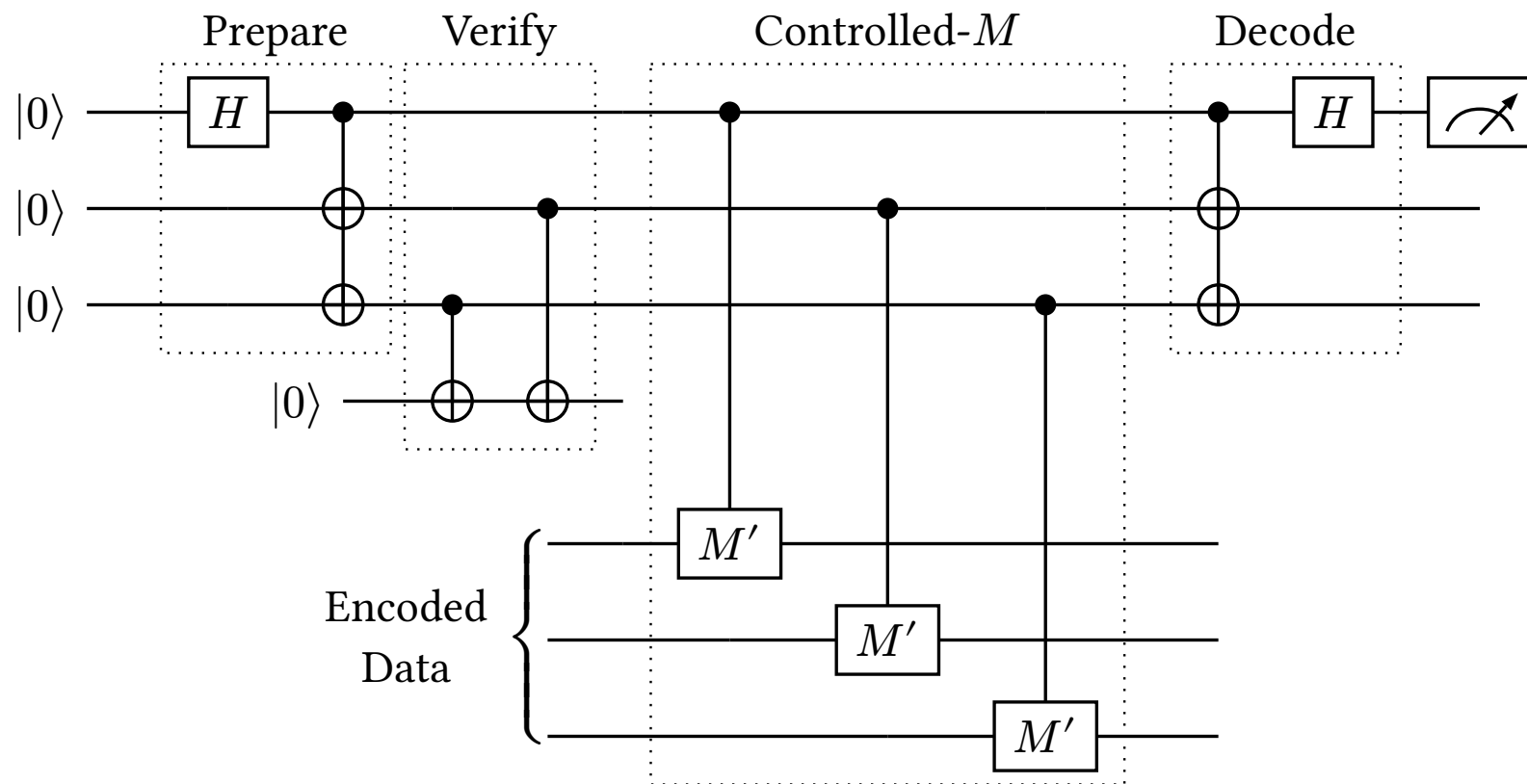
$$R^\mu{}_\nu - \frac{1}{2} R g^\mu{}_\nu = 8\pi G T^\mu{}_\nu$$

```
#import "@preview/quill:0.7.2": *
```

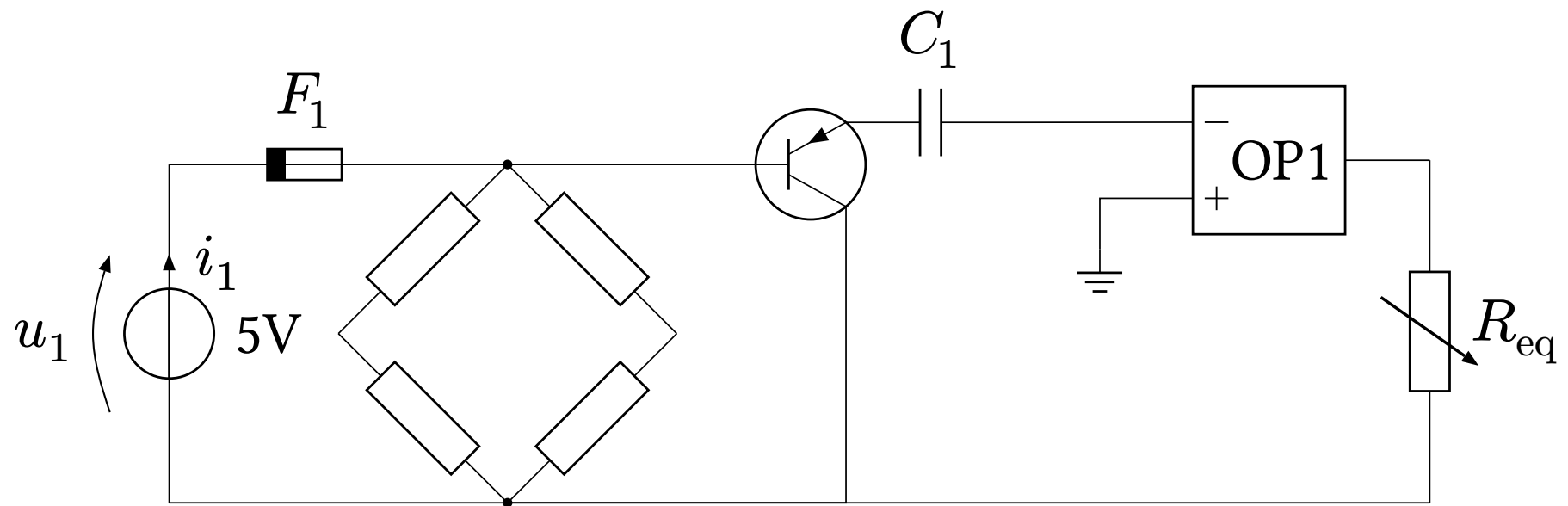
```
1 #import "@preview/quill:0.7.2": *
2 #import "@preview/physica:0.9.8": ket
3
4 #let super = $ (ket(00) + ket(11))/sqrt(2) $
5 #quantum-circuit(
6   lstick($ket(0)$), $H$, ctrl(1), rstick(super, n: 2), [\ ],
7   lstick($ket(0)$), 1, targ(), 1
8 )
```



<https://github.com/Mc-Zen/quill/blob/v0.6.1/examples/fault-tolerant-measurement.typ>



<https://github.com/l0uisgrange/zap/blob/main/examples/example1.typ>

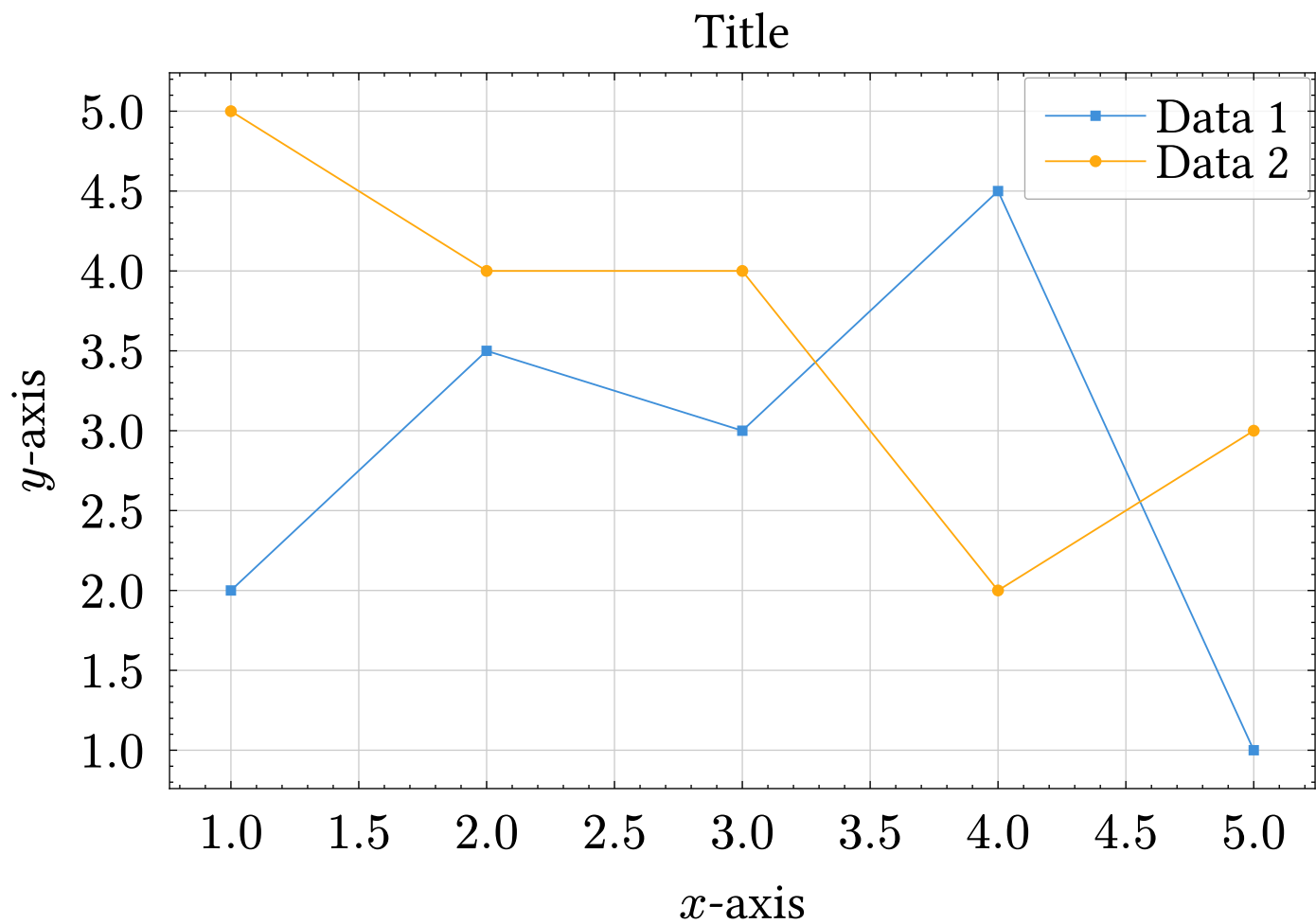


```
#import "@preview/lilaq:0.6.0" as lq
```

```
1 #let data = csv("example.csv").slice(1, none)
2 #let datax = data.map(row => float(row.at(0)))
3 #let datay1 = data.map(row => float(row.at(1)))
4 #let datay2 = data.map(row => float(row.at(2)))
5
6 #lq.diagram(
7   title: [Title],
8   xlabel: [$x$-axis],
9   ylabel: [$y$-axis],
10  width: 16cm,
11  height: 10cm,
12  lq.plot(datax, datay1, mark: "s", label: [Data 1]),
13  lq.plot(datax, datay2, mark: "o", label: [Data 2])
14 )
```

example.csv

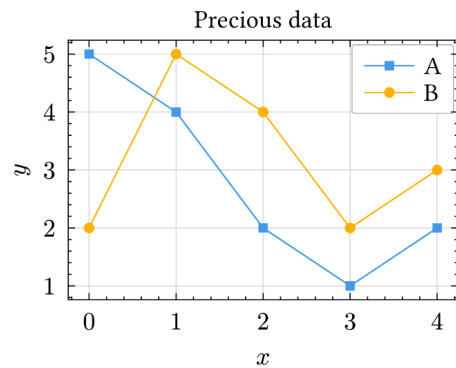
datax	datay1	datay2
1	2	5
2	3.5	4
3	3	4
4	4.5	2
5	1	3



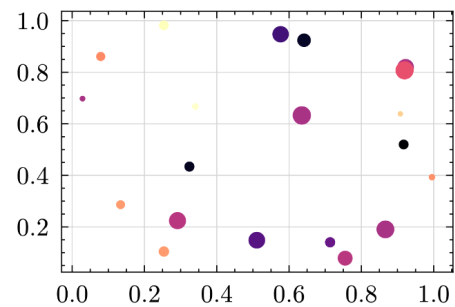
example.csv

datax	datay1	datay2
1	2	5
2	3.5	4
3	3	4
4	4.5	2
5	1	3

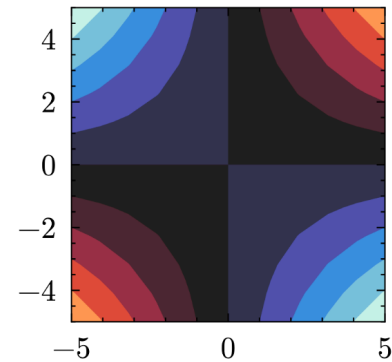
Line plots



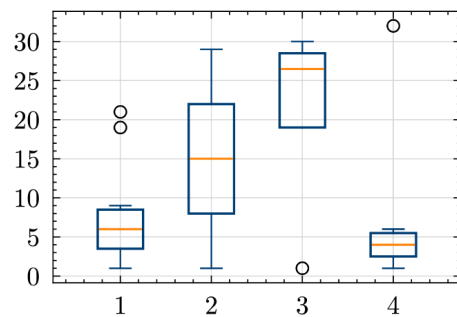
Scatter



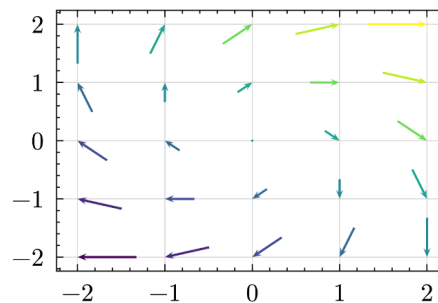
Contour



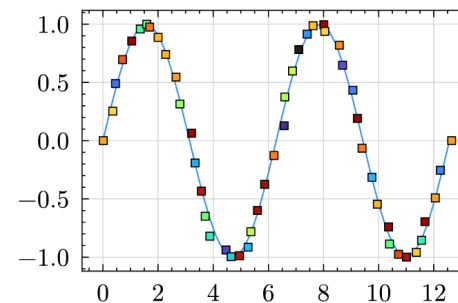
Boxplot



Quiver

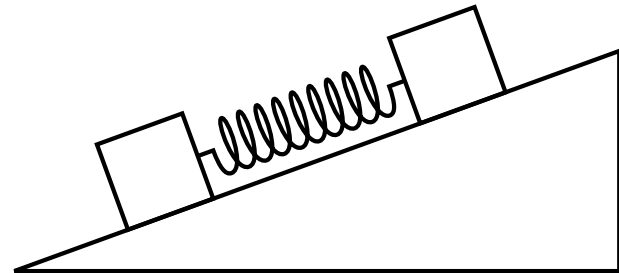


Styled marks

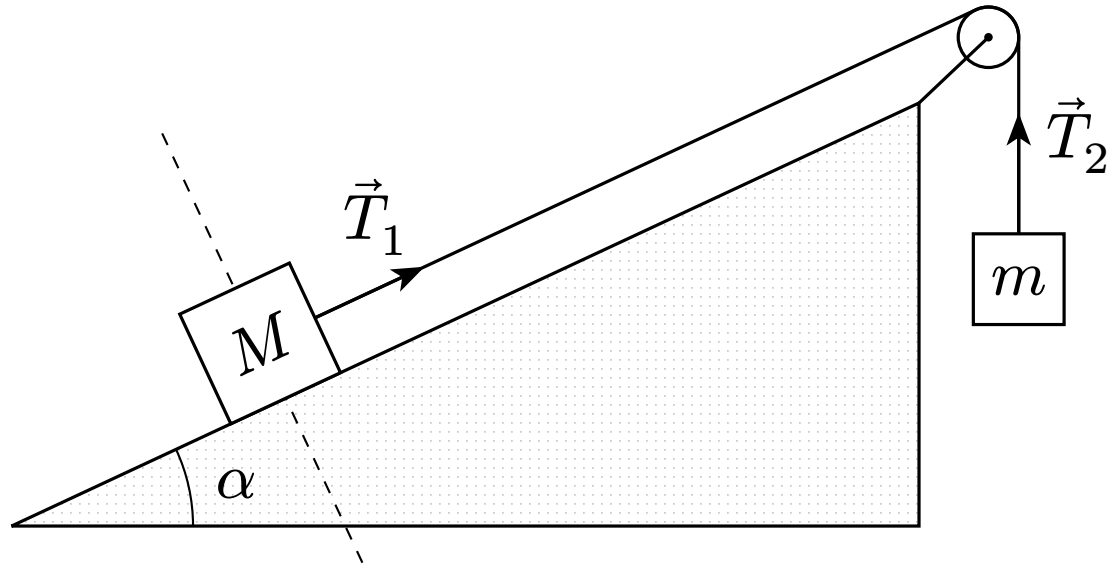


```
#import "@preview/patastrac:0.5.0"
```

```
1  #import "@preview/patastrac:0.5.0"  
2  #patastrac.cetz.canvas({  
3      import patastrac: *  
4      let draw = cetz.standard()  
5      let floor = incline(100, 20deg)  
6      let A = rect(15, 15)  
7      let B = rect(15, 15)  
8      A = stick(A("bl"), floor("tl"))  
9      B = stick(B("br"), floor("tr"))  
10     A = slide(A("c"), +20, 0)  
11     B = slide(B("c"), -20, 0)  
12     let k = spring(A("r"), B("l"))  
13     draw(floor, A, B, k)  
14 })
```



<https://github.com/ZaninDavide/patatrac/blob/main/gallery/incline.typ>



```
#import "@preview/beam:0.1.1"
```

```
1  #import "@preview/beam:0.1.1"  
2  #beam.setup({  
3    import beam: *  
4    laser("laser", (0, 0))  
5    lens("l1", (3, 0))  
6    detector("cam", (4, 0))  
7    beam("", "laser", "l1")  
8    focus("", "l1", "cam")  
9  })
```



```
#import "@preview/wicked:0.2.0": wick
```

```
1 $ :
2 wick(id: #1, macron(Psi))_alpha (x)
3 gamma^mu_(alpha beta)
4 wick(pos: #top, A)_mu (x)
5 wick(Psi)_beta (x)
6 wick(macron(Psi))_eta (y)
7 gamma^nu_(eta rho) (y)
8 wick(pos: #top, A)_nu
9 wick(id: #1, Psi)_rho (y)
10 : $
```

$$: \bar{\Psi}_{\alpha}(x) \gamma_{\alpha\beta}^{\mu} A_{\mu}(x) \underbrace{\Psi_{\beta}(x) \bar{\Psi}_{\eta}(y)} \gamma_{\eta\rho}^{\nu}(y) A_{\nu} \Psi_{\rho}(y) :$$

context

introspection

```
1 | #context { text.size },
2 | #context { text.font }
```

```
1 | Introduction is at:\
2 | #context {
3 |     locate(<intro>).position()
4 | }
5 |
6 | = Introduction <intro>
```

```
20pt, ("libertinus serif", "noto color
emoji")
```

Introduction is at:
(page: 46, x: 430.94pt, y: 211.95pt)

Introduction

```
1 #context {  
2   for h in query(  
3     heading.where(level: 1)  
4   ) {  
5     h.body  
6     linebreak()  
7   }  
8 }
```

Introduction to Typst

Introduction

Bibliography

Introduction

In practice: comparison with LaTeX

Ex 1

Ex 2

Ex 3

boxes

```
1 This is a
2 #box(
3     fill: red, radius: 10pt,
4     inset: (x: 10pt), outset: (y: 10pt),
5     text(fill: white)[`BOX`]
6 )
7
8 This is a
9 #block(fill: blue, radius: 10pt, outset: 10pt,
10     text(fill: white)[`BLOCK`]
11 )
```

This is a 

This is a



2D input in math mode

```
1 | #let func(..args) = repr(args)
2 | $ func(1,2,3; 4,5,6) $
```

arguments(([1], [2], [3]), ([4], [5], [6]))

metadata

```
1 Metadata is invisible content:  
  #metadata("Useful information")  
2  
3 That we can query: #context {  
4   query(metadata).last().value  
5 }
```

Metadata is invisible content:

That we can query: Useful information

WASM plugins

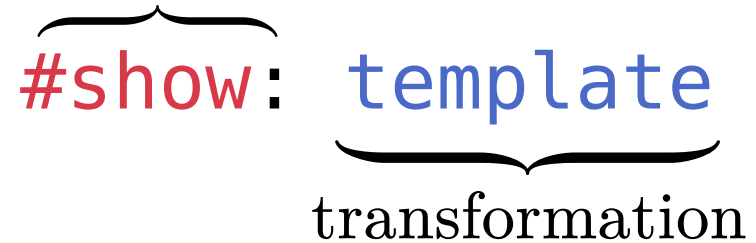
```
1 #let square(n) = {
2   import plugin("example.wasm"): square
3   int.from-bytes(square(n.to-bytes()))
4 }
5 #square(15)
```

225

```
use wasm_minimal_protocol::*;
initiate_protocol!();
#[wasm_func]
pub fn square(bytes: &[u8]) -> Vec<u8> {
    // bytes conversion and squaring
}
```

Templates

select all content below


#show: **template**
transformation

```

1 #import "@preview/unequivocal-ams:0.1.2": *
2 #show: ams-article.with(
3   title: [Knots, Braids and Chaos],
4   authors: (
5     (
6       name: "Desmond Gantry",
7       department: [Department of Mathematics],
8       organization: [University of Rummidge],
9       location: [Rummidge RU15 2TT, United Kingdom],
10      email: "d.gantry@rummidge.ac.uk",
11      url: "www.rummidge.ac.uk/~dp01"
12    ),
13  ),
14  abstract: lorem(100),
15 )
16
17 = Introduction
18 #lorem(80)
19
20 #theorem(lorem(30))
21 #proof(lorem(20))

```

KNOTS, BRAIDS AND CHAOS

DESMOND GANTRY

ABSTRACT. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

1. INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

Theorem 1.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat. $\square$

In practice

Comparison with LaTeX

Installation

Typst

- 1 file
- 50MB
- Packages downloaded as necessary

TeX Live

- 200K files
- 7GB
- Packages preinstalled

Output quality

You know LaTeX.

Typst sample from a [bachelor thesis](#):

Paragraph building and microtypography

Typst

- ✓ Knuth-Plass line breaking
- ✓ Ligatures
- ✓ Protrusions (basic)
- ✓ Character-level justification

2 Density Functional Theory (DFT)

Density Functional Theory (DFT) replaces the many-electron wave function ψ with the electron density ρ as the central quantity to minimize over. This makes the high-dimensional problem of solving the Schrödinger equation for N electrons in $3N$ dimensions tractable by reducing it to a problem of optimizing a function of three spatial variables. In this chapter, we introduce the density, formulate the energy as a functional of that density within the Hohenberg-Kohn theorems, and derive the Kohn-Sham framework used in modern electronic-structure calculations.

2.1 Notation

Throughout this chapter:

- We consider a system of N electrons indexed by $i \in \{1, \dots, N\}$.
- By $\mathbf{r}_i \in \mathbb{R}^3$, we denote the electron positions.
- Let $\psi(\mathbf{r}_1, \dots, \mathbf{r}_N) \in \mathcal{H}$ be the N -electron wave function in Hilbert space (ignoring spin) $\mathcal{H} = \bigwedge_{i=1}^N L^2(\mathbb{R}^3)$ subject to normalization

$$\langle \psi | \psi \rangle \stackrel{!}{=} 1, \quad \text{i.e.} \quad \int_{\mathbb{R}^{3N}} |\psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2 d\mathbf{r}_1 \cdots d\mathbf{r}_N \stackrel{!}{=} 1 \quad (1)$$

- We assume a *non-degenerate* ground state ψ_0 .
- We use atomic units where $\hbar = m_e = e = 1$ and $\frac{1}{4\pi\epsilon_0} = 1$.

2.2 Electron Density

Definition 2.1 (Density): The **density operator** $\hat{\rho}$ for point $\mathbf{r} \in \mathbb{R}^3$ is given by

$$\hat{\rho}(\mathbf{r}) := \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \quad (2)$$

The expectation value of the density operator is called the **electron density**

$$\rho(\mathbf{r}) : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad \rho(\mathbf{r}) := \langle \psi | \hat{\rho}(\mathbf{r}) | \psi \rangle \quad (3)$$

We can explicitly calculate the density by plugging (2) into (3):

$$\rho(\mathbf{r}) := \langle \psi | \hat{\rho}(\mathbf{r}) | \psi \rangle = \int_{\mathbb{R}^{3N}} \psi^*(\mathbf{r}_1, \dots, \mathbf{r}_N) \left(\sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \right) \psi(\mathbf{r}_1, \dots, \mathbf{r}_N) d\mathbf{r}_1 \cdots d\mathbf{r}_N \quad (4)$$

$$= \sum_{i=1}^N \int_{\mathbb{R}^{3N}} |\psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2 \delta(\mathbf{r} - \mathbf{r}_i) d\mathbf{r}_1 \cdots d\mathbf{r}_N \quad (5)$$

$$= N \int_{\mathbb{R}^{3(N-1)}} |\psi(\mathbf{r}, \mathbf{r}_2, \dots, \mathbf{r}_N)|^2 d\mathbf{r}_2 \cdots d\mathbf{r}_N \quad (6)$$

In the last step, while evaluating the delta distribution, we used the anti-symmetry of the

LaTeX

- ✓ Knuth-Plass line breaking
- ✓ Ligatures
- ✓ Protrusions
- ✓ Font expansion
- ✓ Manual kerning adjustments (I never used it...)

Reproducibility (does it still compile?)

Typst

- Not stable (still in 0.x phase)
- Package use is versioned:
`#import "@preview/zap:0.5.0"`
- Easy to co-install older version

LaTeX

- Core: extremely stable
Packages: not stable
- No versioning of packages:
`\usepackage{circuitikz}`
- Difficult to install older version

Case in point: compiling a very nice [LaTeX presentation from 2011](#)

1. ! LaTeX Error: File `dtklogos.sty' not found.

In TeX Live 2016, the package was renamed to `dtk-logos.sty`

2. Package `texshade', Version 1.28 of 2024/01/10

! LaTeX Error: Command \charge already defined.
Or name \end... illegal, see p.192 of the manual.

No clue from Google, I comment out `\usepackage{texshade}` for now.

3. ! Undefined control sequence.

<argument> \MakeIndex

I comment out all uses of `\MakeIndex`.

```
4. [18] (./talk.vrb
! Missing number, treated as zero.
<to be read again>
          \kern
1.10 \documentclass[
          a4paper,11pt]{article}
```

Apparently a breaking change in listings, the fix is to replace

```
1 | postbreak=\mbox{{\smaller\color{gray}$\hookrightarrow$}}
```

with

```
1 | \newsavebox\postbreakbox
2 | \savebox\postbreakbox{\raisebox{0ex}[0ex][0ex]{\smaller\color{gray}$
   | \hookrightarrow$}}
3 | ...
4 | postbreak=\usebox\postbreakbox
```

5. ! Undefined control sequence.
<recently read> \MikTeX

Probably a breaking change in dtk-logos. I remove all occurrences of \MikTeX.

6. ! Undefined control sequence.
<argument> \tikz [remember picture,overlay]\node
[single arrow,fill=DarkSeaG...
1.47 ...1em] at (current page.center) {pdflatex};}

?

Probably a change in the interaction between TikZ overlays and Beamer uncover.
I comment out.

7. ! Package bytefield Error: Macros \wordgroup_r, \wordgroup_l, \endwordgroup_r,
(bytefield) and \endwordgroup_l no longer exist.

See the bytefield package documentation for explanation.

A breaking change in bytefield. Adapting the code.

8. ! Missing } inserted.

```
<inserted text>  
    }  
l.189 \end{frame}
```

?

No idea even which file is the problem... I give up.

Error messages

Typst

```
error: unclosed delimiter
presentation.typ:678:6
678  #slide[
      ^
```

LaTeX

```
! Missing } inserted.
<inserted text>
}
l.189 \end{frame}

?
```

Security

Compiling that LaTeX presentation required `-shell-escape` !

Typst

- No execution of external programs
- No reading outside project directory

LaTeX

- Can run external programs when compiling with `-shell-escape`
- Allows reading any file

LaTeX can read *any file*...

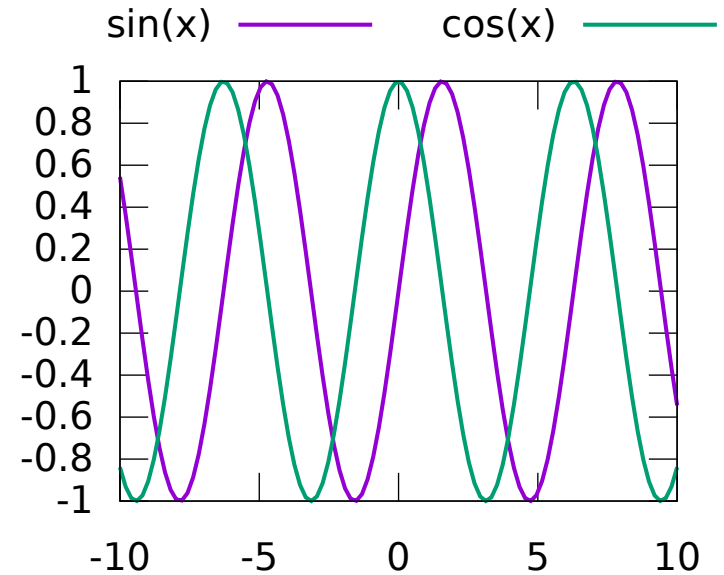
```
1 \documentclass{article}
2 \usepackage{verbatim}
3 \usepackage{xcolor}
4 \begin{document}
5 % Include public key as invisible text in PDF
6 % (would work the same with private key!)
7 \textcolor{white}{\verbatiminput{\string~/.ssh/id_ed25519.pub}}
8 \end{document}
9
```

Typst: no -shell-escape but WebAssembly plugins!

Example: the neoplot package **embeds Gnuplot**

```
1 #import "@preview/neoplot:0.0.4" as gp
2 #show raw.where(lang: "gnuplot"): gp.exec
3
4 ```gnuplot
5 set termoption font "DejaVu Sans,28"
6 set key outside top center horizontal
7 plot sin(x) lw 3, cos(x) lw 3
8 ```
```

- Fairly secure
- Zero installation required
- Works on Windows, macOS, Linux



Syntax

```
1 #slide[Installation][
2   #set text(1.1em)
3   #grid(
4     columns: (50%, 50%),
5     column-gutter: 1em,
6     [
7       *Typst*
8       - 1 file
9       - 50MB
10    ],
11    ...
12  )
13 ]
```

```
1 \begin{frame}{Installation}
2 \fontsize{11pt}{13pt}\selectfont
3 \begin{columns}
4   \begin{column}{0.48\textwidth}
5     \textbf{Typst}
6     \begin{itemize}
7       \item 1 file
8       \item 50MB
9     \end{itemize}
10   \end{column}
11   ...
12 \end{columns}
13 \end{frame}
```

Typst

```
1 | $ \mathbb{Q} = \{ p/q : p \in \mathbb{Z}, q \in \mathbb{N} \} $
```

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\}$$

LaTeX

```
1 | \documentclass{article}
2 | \usepackage{amssymb}
3 | \begin{document}
4 | \[ \mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\} \]
5 | \end{document}
```

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\}$$

Performance

Typst

- Incremental compiler

LaTeX

- Multipass compiler

With my linear algebra lecture notes (140 pages):

Typst

- 6s
- 1 output file
- on change: **instant preview**

LaTeX

- 17s
- 21 output files
- on change: full recompile (6s if lucky)

Programming

Let's print all prime numbers under 50, comma-separated:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

Compare implementations for the **sieve of Eratosthenes** in

1. Plain LaTeX
2. LaTeX3
3. LuaLaTeX
4. Typst

Plain LaTeX

```
1 \documentclass{article}
2 \makeatletter
3 \newcommand{\@crossout}[3]{%
4   \ifnum#1>#3\relax
5   \else
6     \expandafter\gdef\csname
7       isprime@\number#1\endcsname{F}%
8     \expandafter\@crossout\expandafter
9       {\number\numexpr#1+#2\relax}{#2}{#3}%
10   \fi
11 }
12 \newcommand{\sieve}[1]{%
13   \count1=2
14   \loop
15     \expandafter\gdef\csname
16       isprime@\the\count1\endcsname{T}%
17     \ifnum\count1<#1 \advance\count1 by 1 \repeat
18   \count1=2
19   \loop
20     \expandafter\if T\csname
21       isprime@\the\count1\endcsname
22 \edef\tempstart{\number\numexpr\count1*\count1\relax}%
23   \edef\tempstep{\the\count1}%
24   \expandafter\@crossout\expandafter
25     {\tempstart}{\tempstep}{#1}%
26   \fi
27   \ifnum\numexpr\count1*\count1\relax<#1
28     \advance\count1 by 1
29   \repeat
30   \count1=2
31   \def\prime@sep{}%
32   \loop
33     \expandafter\if T\csname
34       isprime@\the\count1\endcsname
35     \prime@sep\the\count1 %
36     \gdef\prime@sep{, }%
37   \fi
38   \ifnum\count1<#1 \advance\count1 by 1 \repeat
39 }
40 \makeatother
41 \begin{document}
42 \sieve{50}
43 \end{document}
```

LaTeX3

```
1 \documentclass{article}
2 \ExplSyntaxOn
3 \intarray_new:Nn \isprime {10001}
4 \seq_new:N \l_sieve_results_seq
5 \NewDocumentCommand{\sieve}{m}{
6   \seq_clear:N \l_sieve_results_seq
7   \int_step_inline:nnnn{2}{1}{#1}{
8     \intarray_gset:Nnn\isprime{##1}{1}
9   }
10  \int_step_inline:nnnn{2}{1}{
11    \fp_to_int:n{floor(sqrt(#1))}}{
12    \int_compare:nNnT{
13      \intarray_item:Nn\isprime{##1}}={1}{
14        \int_step_variable:nnnNn{##1*##1}{##1}{#1}
15        \l_tmp_int{
16          \intarray_gset:Nnn\isprime{\l_tmp_int}{0}
17        }
18      }
19    }
20  \int_step_inline:nnnn{2}{1}{#1}{
21    \int_compare:nNnT{
22      \intarray_item:Nn\isprime{##1}}={1}{
```

```
23      \seq_put_right:Nn\l_sieve_results_seq{##1}
24    }
25  }
26  \seq_use:Nn\l_sieve_results_seq{,~}
27 }
28 \ExplSyntaxOff
29
30 \begin{document}
31 \sieve{50}
32 \end{document}
```

LuaLateX

```
1 \documentclass{article}
2 \usepackage{luacode}
3 \begin{luacode*}
4   function sieve_lua(n)
5     local is_prime = {}
6     for i = 2, n do is_prime[i] = true end
7     for i = 2, math.sqrt(n) do
8       if is_prime[i] then
9         for j = i * i, n, i do
10           is_prime[j] = false
11         end
12       end
13     end
14     local results = {}
15     for i = 2, n do
16       if is_prime[i] then
17         table.insert(results, i)
18       end
19     end
20     tex.print(table.concat(results, ", "))
21   end
22 \end{luacode*}
```

```
23 \newcommand{\sieve}[1]{\directlua{sieve_lua(#1)}}
24
25 \begin{document}
26 \sieve{50}
27 \end{document}
```

Typst

```
1 #let sieve(n) = {  
2   let is-prime = (false, false) + (true,)*(n - 1)  
3   for i in range(2, int(calc.sqrt(n)) + 1) {  
4     if is-prime.at(i) {  
5       for j in range(i * i, n + 1, step: i) {  
6         is-prime.at(j) = false  
7       }  
8     }  
9   }  
10  is-prime.enumerate()  
11    .filter(it => it.at(1))  
12    .map(it => str(it.at(0)))  
13    .join(", ")  
14 }  
15  
16 #sieve(50)
```

Another Typst example: building a point table on the exam's first page

```
1 #context {  
2   let n = query(heading).len()  
3   table(  
4     inset: 1em,  
5     align: center,  
6     columns: (auto,) + (1.5cm,) * n + (auto,),  
7     [Exercise], ..range(n).map(i => str(i + 1)), [Total],  
8     [Points],  
9   ),  
10 }  
11 = Ex 1  
12 = Ex 2  
13 = Ex 3
```

Exercise	1	2	3	Total
Points				

Questions?

Colophon

Slides made in plain Typst (no package needed).

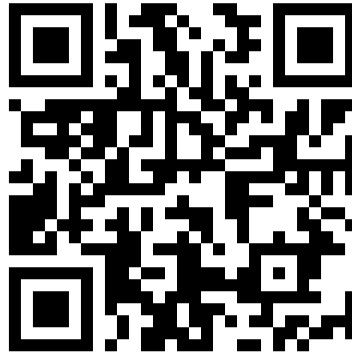
Based on following two presentations:

github.com/sijow/typst-presentation/

github.com/ZaninDavide/typst-intro/

Find this presentation

```
1 | #import "@preview/tiaoma:0.3.0": barcode  
2 | #barcode("https://github.com/ethanc8/typst-intro", "QRCode")
```



<https://github.com/ethanc8/typst-intro>